

Where the Tokens Go: Decomposing and Closing the Decode Gap for a 27B Hybrid LLM on Apple Silicon

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Abstract

Large language models are increasingly run on personal machines, and Apple silicon with unified memory holds a 25–100B-parameter model in one address space. Single-stream decode there is usually called memory-bandwidth bound, yet a widely used runner spends 42 ms per decode step on Qwen3.8-27B in Q4_K on an M2 Ultra, against an estimated 24 ms to stream the active weights at the best rates the GPU achieves. We ask where the rest goes. We decompose a decode step into weight streaming, matvec kernel efficiency, per-dispatch overhead and CPU work between tokens, timing stages with command-buffer GPU timestamps so that sub-millisecond kernels are attributed correctly, and we find that each of the last three costs milliseconds at this scale. Inside DS4, an open-source model-specific C/Metal engine, we add the model and pull three levers, each behind a runtime switch: fewer bytes per token, fewer and better-shaped dispatches, and a shorter CPU path between tokens. Our optimizations shorten the step from 35.0 to 30.8 ms; forward-only, DS4 takes 30.6 ms against llama.cpp’s 42.3, with logits matching llama.cpp at eight decode-step and context-length points. A controlled experiment with the same source checkpoint in Q4_0 and Q4_K shows the block format is worth 8–16% with simple kernels and 2–3% once the large matvecs are fused; MLX reaches a comparable step time on its affine 4-bit conversion. A 2 ms-per-token cost in the sampler, caused by `isfinite` under `-ffast-math`, shows that the CPU gap between tokens must be measured on its own. We report negative results with their mechanisms and release code, switches, logs and scripts.

1 Introduction

Open-weight models in the 20–30B range fit on a laptop or desktop with 64–192 GB of unified memory, and people run them locally for chat, coding agents and private workloads. The question on such a machine is no longer whether a model fits but how fast it decodes, and the standard answer is a roofline argument: a single-stream decode step reads every active weight matrix once, so the step time is bounded below by model bytes over memory bandwidth. For Qwen3.8-27B in Q4_K on an M2 Ultra that is 15.06 GB per step at a nominal 800 GB/s, or 18.8 ms; using the streaming rates we measure for each format on this GPU, about 24.3 ms (§2.3). llama.cpp takes 42.3 ms per decode step (§7). Roughly 18 ms of every step is not accounted for by streaming the weights, and this paper is about where it goes.

That time is hard to see, which is why it persists. A profiler that synchronizes at stage boundaries pays about 0.16 ms per boundary on this machine, more than most of the kernels it is trying to measure, so small kernels vanish into the floor. A system trace reports the GPU 96–98% busy for both llama.cpp and our engine, which sounds as if the roofline were reached; it is not, because a kernel that is running is not necessarily a kernel that is streaming at full rate. And a 2 ms cost per token in the CPU sampler, one fifteenth of the step, is invisible to every GPU-side tool. Our first

contribution is an instrument that makes each of these visible: GPU timestamps of the command buffer for each stage, a CPU inter-token split, and reference logits from llama.cpp at chosen decode steps, so that every change is checked where decode actually runs (§3).

With that instrument we add the model to DS4, an open-source model-specific engine in C and Metal, and pull three levers, each behind a switch so the final build can reproduce any subset (§5): the bytes per step and how fast they move (quantization format and matvec inner loop); the number and shape of dispatches (a layer’s four side projections in one launch, the SwiGLU gate and up in one kernel, each residual add fused into the norm that consumes it, and split-KV attention scaled to the context); and the CPU path between tokens. Switching everything off reproduces the pre-optimization build (35.0 ms per step in the generation-loop definition of §3); everything on gives 30.8 ms, or 30.6 ms forward-only, about 80% of the estimated streaming bound.

Our contributions, and what we learned from each:

1. **A measurement method for Metal decode** with matched timing definitions across engines and a correctness protocol at arbitrary decode steps (§3, §6). The per-dispatch cost it exposes is a first-order term at 27B scale: the original graph issued about 1,070 dispatches per step, comparable in time to the recurrence, the attention and every norm together.
2. **Qwen3.8-27B support and a set of decode optimizations**, each measured in isolation (§5, Figure 3). Their leave-one-out effects are consistent with activation traffic, not occupancy, limiting the Q4_K matvecs: spreading a reduction over more threads did nothing for the wide projections, while reading the activation once per four rows instead of two, as vectors, did.
3. **A controlled quantization-format experiment** (§7.4) with the same source checkpoint in Q4_0 and Q4_K, in two engines and at two kernel-fusion levels. The format is worth 16% in llama.cpp and 8% in DS4 with simple kernels, and 2–3% once the large matvecs are fused; which lever matters depends on how far the kernels are from the streaming bound. MLX, on its affine 4-bit conversion with a smaller output head, reaches a comparable step time from a general framework.
4. **The CPU path as a measured term** (§5.3). A 2 ms sampler cost came from `isfinite` being left as a library call under `-ffast-math`; a first replacement was folded away by the same flags and broke NaN handling. llama.cpp’s own full-vocabulary top- p sampler costs 1.2 ms per token on the same distribution.
5. **Negative results with mechanisms** (§8), a profile of a mixture-of-experts model on the same instrument, and the code, switches, logs and scripts (Appendix B).

DS4 (DwarfStar) is by Salvatore Sanfilippo and contributors [21] (<https://github.com/antirez/ds4>). It implements a handful of architectures as hand-written Metal schedules rather than interpreting a generic GGUF computation graph; by “model-specific” we mean that the per-token dispatch list is written for the model, so fusions that depend on the model or the session are local changes. Our work adds Qwen3.8-27B, including its Gated DeltaNet and gated-attention kernels, and the decode paths and measurements described here; the branches, scripts and logs are at <https://github.com/parasol-aser/ds4>.

Component	Weights	Q4_K bytes
DeltaNet layer ($\times 48$): qkv, z, α, β , out, FFN	383.3M	215.6 MB
Attention layer ($\times 16$): $q gate, k, v$, out, FFN	372.3M	209.4 MB
Decoder layers, from tensor sizes		13.71 GB
Output head, Q8_0 (248320×5120)	1.27B	1.35 GB
Per decode step		15.06 GB

Table 1: Bytes read per decode step for Qwen3.8-27B in pure Q4_K (0.5625 B/weight) with a Q8_0 output head (1.0625 B/weight), from the GGUF tensor sizes. Norms and biases add 20 KB; the auxiliary MTP block (239 MB) and the token-embedding table (715 MB, one row read) are excluded.

2 Background

2.1 Platform and setup

The M-series SoCs share one LPDDR5 pool between CPU and GPU. Metal compiles kernels at runtime and executes them from command buffers; a compute encoder holds many dispatches, which execute in order with memory ordering. There is no host-to-device copy, but the CPU and the GPU share bandwidth, and CPU work between two tokens is on the critical path.

All measurements are on a Mac Studio with an M2 Ultra (24 CPU cores, 60 GPU cores, 192 GB LPDDR5, 800 GB/s nominal), macOS 26.1, Apple clang 17.0.0 (clang-1700.4.4.1). Baselines are llama.cpp at commit `6d9c82ea2bb34e277c0664b8dd3434bfb4dcfb27` (2026-09-09) built with its Metal backend [8], and mlx-lm 0.31.3 on mlx 0.32.2 [17]. The GGUF engines read the same files from the Unsloth release (revision `4ca7207` [22]); MLX uses the community 4-bit conversion (revision `3e6447f`) of the same checkpoint (revision `1d4bf0f` [20]). Appendix B lists file hashes.

2.2 Model

Qwen3.8-27B (GGUF architecture `qwen35`) has 64 decoder layers plus one auxiliary multi-token-prediction block that ordinary decode does not execute; the GGUF therefore has 65 blocks, and the byte accounting below covers the 64 decoder layers, the output norm and head, and one row of the token embedding. In three layers out of four the token mixer is Gated DeltaNet [23]: one fused projection produces q and k (16 heads of 128) and v (48 heads of 128), which pass through a causal depthwise convolution of width 4 followed by SiLU; q and k are L2-normalized per head and q is scaled by $1/\sqrt{128}$; value head h uses query/key head $h \bmod 16$; each value head keeps a 128×128 recurrent state updated by the delta rule with a scalar decay $\exp(A \cdot \text{softplus}(\alpha + b))$, where $A = -\exp(A_{\log})$ is a per-head parameter, b a per-head bias and α comes from a 48-wide projection, and a write strength $\sigma(\beta)$ from a second 48-wide projection; the output is RMS-normalized per head and gated by SiLU(z) from a 6144-wide projection. Every fourth layer is gated grouped-query attention: 24 query heads of dimension 256 with 4 KV heads, per-head RMSNorm on q and k , NEOX RoPE on the first 64 dimensions, and the output multiplied by a sigmoid gate projected alongside q . All layers have a SwiGLU FFN of hidden size 17408; the width is 5120 and the vocabulary has 248,320 entries. Table 1 gives the bytes a decode step reads.

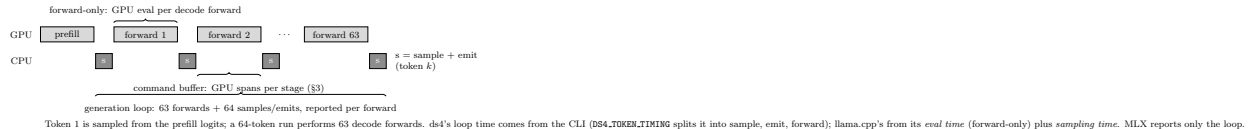


Figure 1: What each reported quantity covers. We report milliseconds per decode forward under two definitions: *forward-only* (the GPU forward of one token) and *generation loop* (forward plus sampling plus emission, divided by the 63 forwards of a 64-token run). Rates in t/s are 1000/ms.

2.3 Streaming bound

At 15.06 GB per step the nominal 800 GB/s bound is 18.8 ms (53 t/s). A more useful reference applies the fastest rate we measure for each format to its bytes: the Q8_0 output-head matvec moves 1.35 GB in 2.01 ms (670 GB/s), and the fastest Q4_K kernel, the fused gate/up projection, moves 100.3 MB in 162.9 μ s (616 GB/s). Then

$$T_{\text{weights}} \approx \frac{13.71 \text{ GB}}{616 \text{ GB/s}} + \frac{1.35 \text{ GB}}{670 \text{ GB/s}} \approx 24.3 \text{ ms},$$

which we call the estimated weight-streaming bound. It assumes every matrix shape streams at the best measured rate for its format and excludes all other work, so it is a bound, not a partition: kernel time below it includes weight traffic, and a slower kernel still spends time moving weights. Throughout, “utilization” of this bound means 24.3 ms divided by the measured step time.

3 Measurement method

Timing definitions. Engines count differently: a 64-token run performs 63 decode forwards, the first token being sampled from the prefill logits, and llama.cpp’s *eval time* excludes sampling while a generation-loop clock includes it. We therefore report milliseconds per decode forward under two explicit definitions (Figure 1). For DS4 the CLI, under DS4_TOKEN_TIMING, reports per token the time in sampling, in emission and in the forward call; for llama.cpp we use its *eval time* per run and add its *sampling time* for the loop; mlx-lm reports only the loop. All headline numbers are means over five runs on an otherwise idle GPU with the model file in the page cache; standard deviations are given in the result set.

Per-stage GPU spans. DS4 can close and reopen its command buffer at named stage boundaries inside a layer under an environment switch. We time each stage with the command buffer’s GPUStartTime and GPUEndTime. Timing the same boundaries with the CPU clock includes a synchronization of about 0.16 ms per boundary, which is larger than the recurrence, the attention or any norm. Isolating stages costs wall-clock time (the instrumented run decodes at 8.7 t/s) but the GPU spans themselves are little affected: their sum, 31.1 ms, is within 2% of the uninstrumented forward (30.6 ms). We use the profile for ranking and for deltas.

System traces. Metal System Trace (xctrace) records GPU busy intervals and command-buffer submissions process-wide; from it we take GPU busy fraction and CPU encode time per token, for DS4 and llama.cpp on the same prompt.

Layer	Stage	Optimizations off		On		On, 8.6k ctx ms/step
		μ s/layer	ms/step	μ s/layer	ms/step	
both	FFN gate/up	195.6	12.52	162.9	10.43	10.44
both	FFN down	112.0	7.16	95.0	6.08	6.07
DeltaNet	projections qkv, z, α, β	115.2	5.53	94.1	4.52	4.48
attention	projections $q gate, k, v$	94.9	1.52	82.1	1.31	1.30
DeltaNet	output norm + out projection	45.3	2.17	48.3	2.32	2.31
attention	gate + out projection	42.8	0.68	44.9	0.72	0.72
DeltaNet	conv + recurrence	23.8	1.14	23.8	1.14	1.14
attention	split-KV attention (+pad, reduce)	106.3	1.70	60.3	0.96	3.36
attention	q/k prologue, V store	7.9	0.13	8.0	0.13	0.14
both	residual adds and norms	29.9	1.91	23.6	1.51	1.48
	output head (Q8.0)		2.01		2.01	2.01
sum of stage spans			36.47		31.13	33.47
uninstrumented step (loop definition)			35.0		30.8	32.6

Table 2: GPU span per stage, Qwen3.8-27B pure Q4_K, M2 Ultra, averaged over decode positions 21–30 of the 20-token prompt (the “8.6k” column over positions 8,591–8,600 of the 8,590-token prompt). “Off” disables the six switches of §5; the single-launch prologue is present in both columns. Each stage runs in its own command buffer; the “both” rows give the per-layer value averaged over all 64 layers.

Reference logits. A small program on the llama.cpp API prefills a prompt, greedy-decodes, and writes the full logit vector at step k , where step 0 is the prefill output and step k is the output after feeding back k generated tokens; DS4 writes the same under a switch. We report mean and maximum absolute difference over the 248,320 entries, the KL divergence of the softmax distributions, and argmax agreement, at steps 0, 4, 16, 40 and 63 on a 20-token prompt, at steps 0, 4 and 63 on a 1,439-token prompt, and at step 4 on an 8,590-token prompt. With greedy decoding on both sides the histories are identical up to k only while the argmax agreed at every earlier step, so we also compare the 64-token greedy outputs directly and report a step only when the histories still agree.

4 Anatomy of a decode step

Table 2 is the per-stage profile with every optimization switched off and on, and Figure 2 groups it. Three observations organize the rest of the paper.

The large matvecs are near, not at, the bound. The FFN gate/up pair moves 100.3 MB per layer. As two ggml-style matvecs plus a SwiGLU kernel its effective rate is 513 GB/s; fused and vectorized (§5), 616 GB/s. The 5120-output projections (FFN down, attention and DeltaNet output) run at 390–450 GB/s with the ggml-style kernel. The Q8.0 output head reaches 670 GB/s on the same memory system. Effective rate is bytes over kernel time, not a measurement of DRAM traffic; the difference between formats is consistent with the ALU cost of unpacking Q4_K’s 6-bit sub-block scales and minimums, and §7.4 tests that directly.

Dispatch count matters at this scale. The pre-optimization graph issued about 1,070 dispatches per step (16 per DeltaNet layer, 19 per attention layer, plus embedding and head). At a few microseconds of ramp and drain each, that is 3–5 ms, as much as the recurrence, the attention and all norms together. The final graph issues about 620.

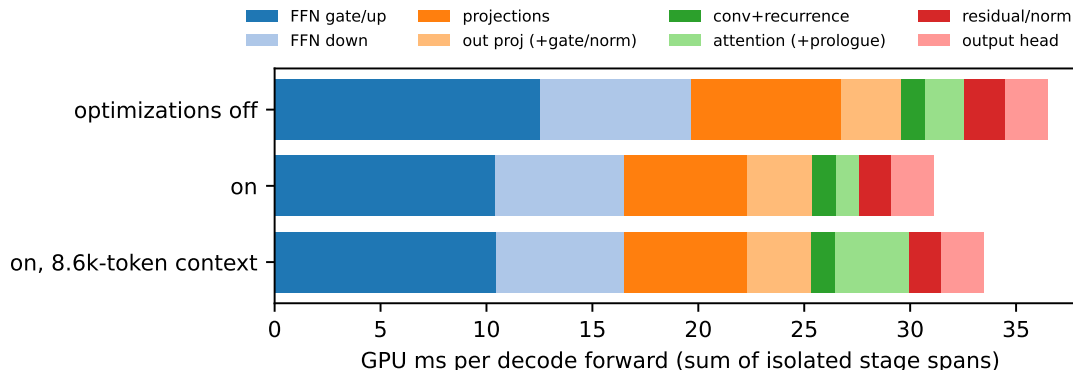


Figure 2: Table 2 grouped by stage type.

GPU busy is not GPU useful. On the mixed-recipe UD-Q4_K_M file, the system trace shows DS4 at 98.3% GPU busy against 95.6% for llama.cpp, with about 1 ms of CPU encode time per token against 4.9 ms. Both engines keep the GPU occupied; the gap to the bound is mostly inside kernels, not between them, which is why the work below is mostly about kernel shape and fusion.

5 Optimizations by lever

Six optimizations have runtime switches (Appendix B lists them); Figure 3 and Table 8 give their leave-one-out effects around the optimized build, in milliseconds per decode forward. Leave-one-out effects around the final build need not equal standalone gains from the original build, and several switches remove the same dispatch overhead, so they do not sum to the all-off difference.

5.1 Bytes: how fast a byte moves

The format fixes the byte count; the kernel decides the effective rate. Two changes to the Q4_K inner loop were worth 9.6% on the fused gate/up kernel (178.6 to 162.9 μ s per layer): staging the 32 activation values a lane needs as eight `float4` loads instead of 32 scalar loads (the activation pointer’s alignment is not provable to the compiler, so it does not vectorize the loads itself), and loading each lane’s two 8-byte nibble runs as `uint2` instead of four `uint16`. Amortizing one activation staging over four rows per simdgroup instead of two helped the 10240- and 12288-output projections in the stage profile and hurt the 5120-output ones, so the row count is chosen per shape; end to end the difference is within noise. A K-split variant, which spreads one row’s reduction over the simdgroups of a threadgroup and combines through threadgroup memory, helps only the long-row FFN down projection (17408 inputs: 112 to 95 μ s) and is used only there.

5.2 Dispatches: fewer and better shaped

Multi-weight matvec. A DeltaNet layer’s four side projections (qkv , z , α , β) share one activation and one input width. One kernel with up to four weight segments, dispatched as a single grid whose threadgroup ranges map to segments, writes the outputs back to back into the existing batch buffer; the convolution and recurrence read z , α and β through views. The same kernel issues $q|gate$, k and v of an attention layer. Four launches become one, and the 48-output α and β heads no longer each pay a dispatch.

Fused gate/up SwiGLU. One kernel streams the gate and up weights against one staged activation and writes $\text{SiLU}(g) \cdot u$; the elementwise pass disappears and the activation is read once per four rows instead of once per two.

Residual add fused into the next norm. Each residual add is fused with the RMSNorm that consumes its result, so a layer’s FFN epilogue produces the next layer’s attention-norm input; two adds and two norms per layer become two kernels.

Single attention prologue. Per-head RMSNorm plus RoPE for the query heads, the same for the key heads into the f16 K cache, and the V store are one grid with three kinds of threadgroup instead of three launches (this one has no switch).

Context-scaled split-KV. The split-KV decode attention kernel adapted from ggml partitions the keys over 32 workgroups and reduces their partials in a second kernel. At a 20-token context 31 workgroups are idle and the reduce dominates. Scaling the workgroup count with the context length, and letting a single workgroup write its normalized output directly, cuts the attention stage from 106 to 60 μs per layer; the shared reduce kernel had silently assumed 32 workgroups and was made correct for fewer.

5.3 The CPU path between tokens

Sampling settings first, since they decide what the sampler has to do. The model’s generation configuration specifies temperature 1.0, top- p 0.95 and top- k 20; llama.cpp reads these from the file’s metadata and its default chain applied top- k 20 and min- p 0.05 in our runs. DS4’s Qwen defaults are temperature 1.0 and top- p 0.95 with no top- k and no min- p , so its sampler scans the full 248,320-entry distribution every token; this is the configuration studied here, and for comparison we also ran llama.cpp with the same settings (`--top-k 0 --min-p 0`).

Under those settings the CLI spent 2.2 ms per token in DS4’s sampler against 30.5 ms in the forward. The cost was not the softmax or the sort but `isfinite()` on each logit, twice per token: the engine is compiled with `-ffast-math`, under which Apple clang 17 leaves `isfinite` as a call to `__isfinitef` and the scan runs scalar at about 4 ns per element. A first replacement, an exponent-bit test on the float value, was folded to “always finite” by the optimizer under the same flags, admitted NaN and $-\infty$ logits, and failed 1,459 unit tests; the working version tests the bits as loaded from memory, which the optimizer does not classify. The finite test alone did not recover the time; the gain came from rewriting the max scan and the top- p pass branch-free and block-wise so they vectorize, with a degree-5 fit of 2^x (relative error under 2×10^{-6}) for the tail mass that cannot enter the 512-entry candidate heap, and a lowering-cutoff fallback instead of a full-vocabulary sort. The sampler now costs 0.41 ms per token. The flags cut both ways: the same sampler built with `-fno-finite-math-only`, which inlines `isfinite`, costs 0.74 ms, because the vectorized scans lose the fast-math assumptions. For scale, llama.cpp’s sampler costs 0.13 ms per token with its default top- k 20 chain and 1.21 ms with the full-vocabulary top- p settings. Two lessons are general: measure the inter-token gap on the CPU, and treat fast-math flags as part of the numerical contract of code that handles non-finite values. Whether other compilers or SDK versions behave the same way we have not checked.

Prompt	step	mean $ \Delta $	max $ \Delta $	KL	argmax
20-token prompt	0	0.0377	0.299	2.3e-05	agrees
20-token prompt	4	0.0042	0.045	1.7e-07	agrees
20-token prompt	16	0.0042	0.045	-7.2e-09	agrees
20-token prompt	40	0.0061	0.055	2.5e-07	agrees
1,439-token prompt	0	0.0900	0.522	4.2e-05	agrees
1,439-token prompt	4	0.0243	0.213	2.0e-06	agrees
1,439-token prompt	63	0.0138	0.199	3.5e-05	agrees
8,590-token prompt	4	0.0153	0.205	5.0e-07	agrees

Table 3: Logit difference from llama.cpp, pure Q4_K, over 248,320 entries; KL is $D(\text{DS4} \parallel \text{llama.cpp})$ of the softmax distributions.

File	bytes/weight	PPL	paired vs pure Q4_K
pure Q4_K, imatrix	0.5625	5.80 ± 0.07	
UD-Q4_K_M	0.58	5.85 ± 0.07	$+0.8\% \pm 0.4$
Q8_0	1.0625	5.86 ± 0.07	$+1.0\% \pm 0.5$
pure Q4_0, imatrix	0.5625	6.03 ± 0.07	$+3.9\% \pm 0.9$
MLX 4-bit affine, group 64	0.5625	5.98	

Table 4: Perplexity on the first 40 chunks of 2048 tokens of wikitext-2 test, scoring positions 1025–2047 of each chunk (**llama-perplexity**; a script with the same chunking for the MLX model). The $\pm$ after each PPL is llama-perplexity’s token-level standard error; the last column is the mean paired difference of per-chunk loss, as a percentage, with its chunk-level standard error. Bytes per weight are for the layer tensors.

6 Correctness

Table 3 compares DS4 with llama.cpp on the same Q4_K file at eight (prompt, step) points where the greedy histories still agree. The argmax agrees at every point and the distributions are close; the 64-token greedy outputs are identical on the Q4_0, UD-Q4_K_M and Q8_0 files and diverge at token 46 on Q4_K, where a near-tie resolves differently (so the step-63 point on the 20-token prompt compares different histories and is omitted). The values were unchanged, to the printed precision, across every optimization added in §5; prefill logits are bit-identical between the initial and final builds because no prefill kernel changed. Two prompts and eight steps are spot checks, not a proof of equivalence over long generations.

6.1 Quality of the files

Because both engines read the same file and agree to this precision, the quality of a file is a property of the quantization, not of the engine. Table 4 gives wikitext-2 perplexity [16], and because all files score the same chunks we report paired differences of the per-chunk loss with a chunk-level standard error, which is the informative statistic. Pure Q4_K, UD-Q4_K_M and Q8_0 are within 1% of each other (the differences are 1–2 chunk-level standard errors, and the ordering is not the one bit width would predict); pure Q4_0 and the MLX conversion are 3–4% worse, at 4 standard errors for Q4_0. Forty chunks of wikitext-2 are a narrow view of quality.

File	DS4 (ms/forward)		llama.cpp (ms/forward)			prefill DS4/llama
	forward	loop	forward	loop, default	loop, matched	
pure Q4_K	30.6	30.7	42.3	42.4	43.6	272 / 269
pure Q4_0	29.9	30.0	35.4	35.5	36.6	289 / 304
UD-Q4_K_M	37.2	37.3	44.1	44.2	45.4	262 / 260
Q8_0	44.1	44.2	49.8	49.9	51.1	305 / 303

Table 5: Milliseconds per decode forward, 20-token ChatML prompt with an empty thinking block, 64 generated tokens, mean of five runs (standard deviations ≤ 0.2 ms). “loop” is the generation loop per forward; DS4’s loop is greedy, llama.cpp’s “default” is its metadata-driven chain (top- k 20, min- p 0.05) and “matched” is DS4’s setting (top- p 0.95, no top- k). Prefill is t/s on the 1,439-token prompt. Rates: DS4 32.6 t/s and llama.cpp 23.7 t/s forward-only on pure Q4_K.

7 Evaluation

7.1 Decode step time

Table 5 gives the headline comparison under both timing definitions. On pure Q4_K, DS4 takes 30.6 ms per decode forward against 42.3 for llama.cpp, a 28% shorter step; the estimated streaming bound of 24.3 ms is 79% of it. The step is about 26 ms of weight-carrying matvecs at 390–616 GB/s effective, 2 ms for the output head, and 2–3 ms of small kernels and per-dispatch ramp. On the mixed-recipe file the step is 16% shorter than llama.cpp’s (only its Q4_K tensors take the fused paths) and on Q8_0 11% (only the format-independent fusions apply). Sampling adds 0.4 ms per token in DS4 and 0.1 ms (default chain) or 1.2 ms (full-vocabulary top- p) in llama.cpp.

The pre-optimization build takes 35.0 ms per forward (generation-loop definition) on pure Q4_K against 30.8 with everything on (Table 8), so the optimizations of §5 shorten the step by 4.2 ms (12%); the remaining difference to llama.cpp’s 42.4 comes from the engine’s existing design, chiefly its shorter per-token dispatch list and CPU encode path.

7.2 Ablation

The fused gate/up kernel is the largest single item, followed by the multi-weight projections; the vectorized single-weight matvec and the K-split overlap because both replace the ggml-style kernel on the FFN down projection. The residual+norm fusion and the context-scaled attention are small at a 20-token context.

7.3 Context length and the MLX baseline

Figure 4 sweeps the prompt from 1.4k to 33k tokens (values in Table 9, appendix; three runs each). DS4’s step grows from 31.4 to 38.1 ms and llama.cpp’s from 42.5 to 49.4. The DeltaNet layers cost the same at every length (Table 2); the growth is the sixteen attention layers reading a K/V cache of 64 KiB per position summed over those layers, plus the split-KV reduce (0.96 to 3.36 ms per step from 20 to 8,590 tokens). Prefill throughput is the same in DS4 and llama.cpp within 4% at every length: prefill is a matrix–matrix workload on comparable dequantize-and-multiply kernels.

MLX is a practical baseline rather than a controlled one. Its community conversion stores the layers in 4-bit affine format (a 16-bit scale and bias per group of 64; 4.5 bits per weight, the same as Q4_K) but also the output head, at 715 MB against our Q8_0 head’s 1.35 GB, about 1 ms per step at these rates; mlx-lm reports only the generation loop; and the model code is Python over the framework’s primitives. On the same 20-token prompt its loop takes 29.4 ms per forward,

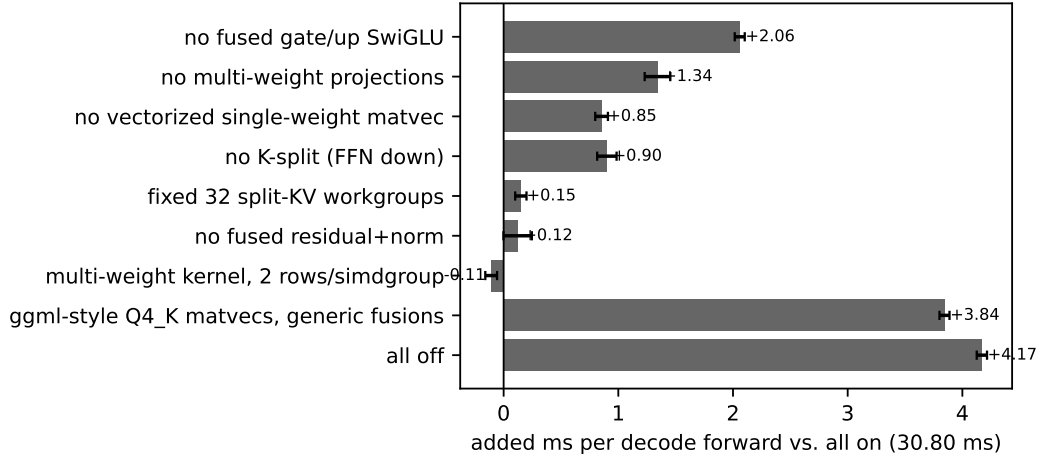


Figure 3: Leave-one-out effect of each switch on the decode step, in milliseconds per forward added relative to the all-on build; three runs each, error bars are the combined standard deviation. Table 8 in the appendix has the values.

Engine	Kernels	Q4_K	Q4_0
llama.cpp	stock (forward-only)	42.3	35.4
DS4	per-format ggml-style matvecs, generic fusions	34.6	31.8
DS4	+ fused gate/up and multi-weight kernels	30.8	30.0
wikitext-2 PPL		5.80	6.03

Table 6: Milliseconds per decode forward for the same source checkpoint in Q4_K and Q4_0 (both pure, same imatrix, Q8_0 head), greedy, 20-token prompt; DS4 rows use the generation-loop definition (forward-only is 0.1 ms less), mean of three to five runs. “Generic fusions” are the residual+norm fusion, the single prologue and the context-scaled attention, which do not depend on the format. Q4_0 logits agree with llama.cpp to a mean $|\Delta|$ of 0.0009 at step 4, and its 64-token greedy output is identical.

comparable to DS4’s 30.7, and it tracks DS4 across context lengths (30.0 to 36.5 ms) with prefill 13–18% slower. The comparison shows that a general framework on a cheaper format reaches a similar step time; it does not isolate the format, which is what §7.4 does.

7.4 The format lever: Q4_0 against Q4_K

To separate the quantization format from everything else we built two files from the same Q8_0 source with the same importance matrix and the same Q8_0 output head: pure Q4_K and pure Q4_0, the same bit budget (16.0 GB on disk) with different block layouts and therefore different reconstructed values. Q4_0 stores one 16-bit scale per 32 weights and dequantizes with one multiply-add per weight; Q4_K packs 6-bit sub-block scales and minimums into a 256-weight super-block that must be unpacked. We ported the ggml-style Q4_0 matvec into DS4 and then the fused gate/up and multi-weight kernels; Table 6 gives the step time at each kernel level in both engines.

With simple kernels the format alone shortens the step by 16% in llama.cpp and 8% in DS4: the Q4_0 gate/up pair takes 171 μ s per layer against 196 for Q4_K. With the fused kernels the

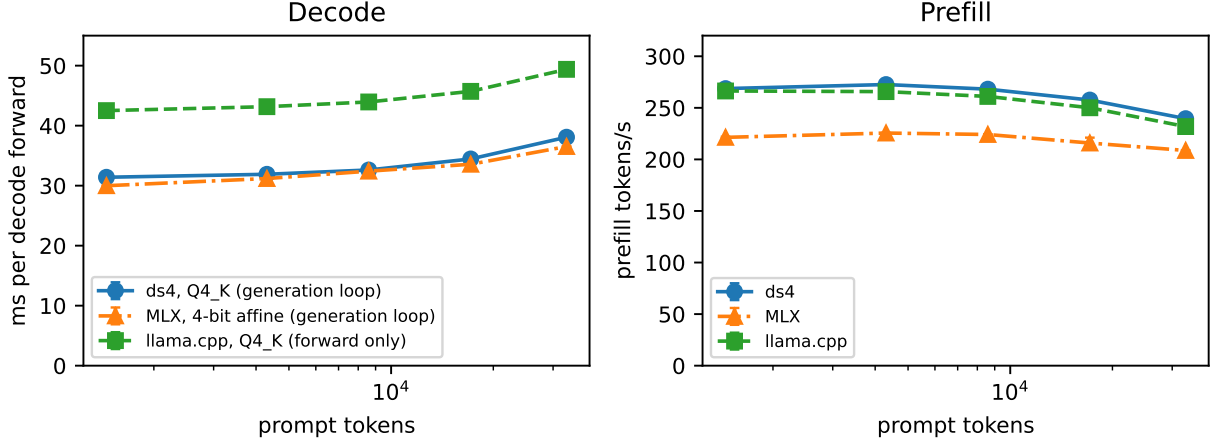


Figure 4: Decode step and prefill throughput versus prompt length, three runs each. DS4 and llama.cpp on pure Q4_K; MLX on its 4-bit affine conversion (generation loop, 4-bit head).

two formats land at 30.0 and 30.8 ms, 2.5% apart. The stage profile is consistent with a simple explanation: the fused gate/up kernel takes 166 μ s in Q4_0 and 163 in Q4_K, both about 610 GB/s effective, so on the largest projection the cheaper dequantization has nothing left to buy once the kernel is fused; Q4_0 still gains on the narrower projections (84 against 94 μ s for the DeltaNet projections, 75 against 82 for attention) and loses on the FFN down projection (103 against 95), which has a K-split kernel only for Q4_K. The format lever matters in proportion to how far the kernels are from the streaming bound. It costs about 4% in perplexity, the same trade the MLX conversion makes.

7.5 A second family: mixture of experts

DS4 was written for DeepSeek V4 Flash, a mixture-of-experts model, and its decode there is shaped by a different lever: only the routed experts of a token are read, so bytes per step depend on routing, and the engine gathers expert weights from a combined buffer and fuses the router with the selection. On the 86 GB mixed-recipe file (IQ2.XXS routed experts, Q8_0 attention projections, shared experts and head), DS4 takes 33.9 ms per decode forward (loop definition) against 39.6 forward-only for llama.cpp on the same file (greedy, short prompt). Table 7 is the same profiler on this model. The routed experts, the query path and the attention output projection account for two thirds of the step, and no single projection dominates the way the FFN does in the dense model; the decomposition applies per family with a different dominant term, and the specific kernels do not transfer.

8 What did not work

K-split on wide matrices (measured). Splitting the reduction across the simdgroups of a threadgroup multiplies the thread count for a fixed number of output rows. It helped only the long-row FFN down projection and was neutral or slightly worse on the 10240- and 12288-output projections. This is consistent with activation re-reads per row, not occupancy, being the limiter on those shapes, since rows per simdgroup was the parameter that moved them; a change of rows per simdgroup alters several things at once, and we did not isolate the mechanism further.

Profiler stage (43 layers)	$\mu\text{s}/\text{layer}$	ms/step
routed_moe	202	8.68
q_path	165	7.10
attn_inv_rope	134	5.77
attn_output	118	5.09
router	51	2.19
kv_path	33	1.41
attn_hc_pre	27	1.14
ffn_hc_pre	27	1.15
shared_down	23	1.00
other (compressor and indexer updates, norms; amortized)		0.58
sum of stage spans		34.1

Table 7: Per-stage GPU span for DeepSeek V4 Flash decode on DS4, short prompt, decode positions 18–27, using the engine’s stage labels. A stage whose kernels are encoded in a neighbour’s command buffer is attributed to that neighbour, which is why some labels report zero and are omitted.

Output gate fused into the attention reduce (measured). A copy of the split-KV reduce that multiplies by $\sigma(\text{gate})$ before its single store was consistently 20 μs per layer slower than the shared reduce followed by a trivial elementwise gate kernel, even with the gate arithmetic removed. The copy specializes its loops on function constants where the shared kernel takes them as runtime arguments; we did not confirm that this is the cause, and dropped the fusion.

Convolution folded into the recurrence (rejected by estimate, not measured). In the proposed design each of the 1,536 recurrence threadgroups would recompute the causal convolution for the 256 q/k channels of its head; with 16 distinct q/k heads shared by 48 value heads, each unique channel would be recomputed 96 times, and the convolution state re-read as often. We judged the redundant work larger than the one dispatch it would save and did not implement it.

9 Discussion

What the levers are worth here. Of the 11.6 ms between llama.cpp’s step and DS4’s, 4.2 ms are the optimizations of this paper and the rest is the engine’s existing design; the controlled experiments put the format lever at 0.8 ms once kernels are fused and at 2.8–6.9 ms when they are not. GPU timestamps, CPU critical-path effects and context-dependent attention partitioning are not unique to unified memory; what is specific to this setting is that at 27B scale on one GPU, per-dispatch cost and CPU gaps are each a few percent of the step and add up, and that a slow sampler is invisible to every GPU-side tool.

Hand-written schedules against frameworks. DS4 implements each family as a hand-written Metal schedule; the cost is that each new family is a port, the benefit is that fusions which depend on the model or the session (batching a layer’s side projections, scaling split-KV work to the context) are local changes with measured effects. llama.cpp and MLX both fuse some operators; we have not audited which, and the MLX result shows that a cheaper format on a general framework reaches a similar step time.

What is left. The remaining 20% to the estimated bound is split between the latency-bound recurrence, the attention path and per-dispatch ramp; §7.4 indicates that dequantization is a small term once the large matvecs are fused. The model ships a multi-token prediction block, so several candidate positions could be verified per weight pass; the benefit depends on acceptance rates, drafting cost and verifying the recurrent state, and we have not measured it.

10 Related work

Local inference runtimes. llama.cpp and ggml [8, 7] are the reference for consumer-hardware inference; DS4’s Q4_K and Q4_0 matvecs and its split-KV decode attention are ports and adaptations of theirs, and the fused kernels of §5 are new. MLX [10] is Apple’s array framework; mlx-lm [17] implements the Gated DeltaNet families.

Roofline and phase analyses. FlashAttention [3] made attention IO-aware for training and prefill; Flash-Decoding [4] partitions decode attention along the key sequence, which is the design the split-KV kernel here follows; disaggregated serving [18, 25] separates the prefill and decode phases at the cluster level. These target discrete GPUs and batched serving; we measure single-stream decode where the CPU shares the memory system.

Hybrid linear-attention models. Gated DeltaNet [23] and the Qwen3-Next and Qwen3.5 families [19, 20] interleave linear-attention layers with a minority of softmax-attention layers; the reference CUDA kernels [24] cannot be used on Metal, and the decode recurrence here is a direct implementation of the delta rule.

Quantization formats. Weight-only quantization methods [6, 15], the k-quant block formats [11], importance-matrix calibration [12] and mixed-recipe files [22] trade bytes for dequantization work; §7.4 measures that trade on an Apple GPU.

Speculative and multi-token decoding. Verifying several tokens per weight pass [13, 2, 1, 14] is the standard route past the streaming bound; the model used here ships a multi-token prediction block [9, 5].

11 Limitations

All results are from one machine, an M2 Ultra with a 60-core GPU; achieved rates and the balance between levers will differ on other generations (the M5 family adds GPU matrix units that change the prefill regime), and a device sweep is the obvious extension. The decomposition is for one dense hybrid family; the MoE result is a step time and a profile, not an ablation. The per-stage profiler perturbs what it measures by isolating stages, so we use it for rankings and deltas. Effective bandwidths are bytes over kernel time, not counter measurements; the mechanisms proposed for the kernel results are consistent with the data but were not isolated with microbenchmarks. The correctness checks are spot checks at eight points; perplexity is on forty chunks of one corpus; the sampler finding depends on the compiler and flags.

12 Conclusion

A decode step on Apple silicon is not simply bandwidth bound at the 27B scale: matvec kernel shape, dispatch count and CPU work between tokens each cost milliseconds, and each needs its own measurement. GPU timestamps per stage and a CPU inter-token split turned each optimization into a targeted change with a measured effect, the switches make the account auditable, and the controlled format experiment shows where one lever stops and another starts.

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A Additional results

A.1 Downstream check

As a coarse check on reasoning tasks we ran DS4’s built-in evaluation harness on its first ten questions (drawn from GPQA Diamond, SuperGPQA and AIME 2025) with thinking enabled, greedy decoding and a 2,500-token reply budget. The harness forces the thinking block closed near the budget and extracts an answer from what follows, so a cut-off answer can still be correct. Pure Q4_K scored 8/10 (3 cut off), pure Q4.0 7/10 (3), Q8.0 7/10 (4): within one question of each other, which is the resolution of ten items, and not evidence of a ranking.

Configuration	ms/forward	Δ vs all on
all optimizations on	30.80 ± 0.04	
no fused gate/up SwiGLU	32.86 ± 0.01	+2.06
no multi-weight projections	32.14 ± 0.10	+1.34
no vectorized single-weight matvec	31.65 ± 0.03	+0.85
no K-split (FFN down)	31.70 ± 0.07	+0.90
fixed 32 split-KV workgroups	30.95 ± 0.02	+0.15
no fused residual+norm	30.92 ± 0.12	+0.12
multi-weight kernel, 2 rows/simdgroup	30.69 ± 0.03	-0.11
ggml-style Q4_K matvecs, generic fusions	34.64 ± 0.01	+3.84
all off	34.97 ± 0.01	+4.17

Table 8: Leave-one-out ablation, pure Q4_K, greedy, 20-token prompt, 64 tokens, generation loop per forward, mean of three runs.

Prompt tokens	ms per decode forward			prefill t/s		
	DS4 loop	MLX loop	llama.cpp fwd	DS4	MLX	llama.cpp
1450	31.4 ± 0.1	30.0 ± 0.0	42.5 ± 0.0	269 ± 0	221 ± 1	266 ± 0
4306	31.9 ± 0.1	31.2 ± 0.0	43.2 ± 0.0	273 ± 0	226 ± 0	266 ± 0
8590	32.6 ± 0.1	32.4 ± 0.0	43.9 ± 0.0	268 ± 0	224 ± 0	261 ± 0
17158	34.5 ± 0.0	33.6 ± 0.0	45.7 ± 0.1	258 ± 0	216 ± 5	250 ± 0
32866	38.1 ± 0.1	36.5 ± 0.0	49.4 ± 0.0	240 ± 0	209 ± 0	232 ± 0

Table 9: Context-length sweep, greedy, 64 decoded tokens, mean $\pm$ sd of three runs. DS4 and llama.cpp on pure Q4_K, MLX on its 4-bit affine conversion; token counts are llama.cpp’s for the shared prompt files (MLX’s tokenizer counts one fewer).

B Reproduction

The code is our fork of the DS4 repository [21] at <https://github.com/parasol-asar/ds4>. Three branches build on each other: **general-decode-fixes** (sampler and diagnostics), **qwen38-support** (the model and its decode paths) and **paper-artifact**, which holds this document, the scripts (**paper/scripts/**), the raw logs (**paper/logs/**) and the result set (**results.json**) from which the tables and figures are generated. Everything below was run on macOS 26.1 with Xcode’s command-line tools; a llama.cpp checkout is needed for the baselines, the reference logits and the quantization. Every command is run from the repository root unless stated.

1. Build

```
git clone https://github.com/parasol-asar/ds4 && cd ds4 && git checkout paper-artifact
make                                     # ds4, ds4-server, ds4-agent, ds4-eval
git clone https://github.com/ggml-org/llama.cpp ~/github/llama.cpp
(cd ~/github/llama.cpp && git checkout 6d9c82ea2bb34e277c0664b8dd3434bfb4dcfb27 \
    && cmake -B build && cmake --build build -j)
pip3 install mlx==0.32.2 mlx-lm==0.31.3 huggingface_hub matplotlib numpy
```

2. Models

```

./download_model.sh qwen38-q8          # Qwen3.8-27B-Q8_0.gguf      (unsloth/Qwen3.8-27B-GGUF @ 4
    ca7207)
./download_model.sh qwen38-q4km        # Qwen3.8-27B-UD-Q4_K_M.gguf (same release)
./download_model.sh ds4f-q2           # DeepSeek V4 Flash mixed-recipe file (-0731 build)
paper/scripts/run_q40.sh               # fetches imatrix_unsloth.gguf, builds the pure Q4_K and
    Q4_0 files, prints SHA-256

```

SHA-256 of the files we measured (paper/logs/sha256.txt):

File	SHA-256 (first 16 and last 8 hex digits)
Qwen3.8-27B-Q4_K.gguf	7d4154def38abac7...af9a9526
Qwen3.8-27B-Q4_0.gguf	129e216a45980381...0fc7a34e
Qwen3.8-27B-UD-Q4_K_M.gguf	322e194ff79741c7...39123482
Qwen3.8-27B-Q8_0.gguf	a680f44a06920e5d...9b67e348
imatrix_unsloth.gguf	0ee5b10bd0c2fa21...6599f1c1

The DeepSeek file we profiled is an earlier build of the `ds4f-q2` recipe (same name without the `-0731` suffix) that is no longer distributed; its step time in §7 should be re-measured on the current file.

3. Experiments

```

cd paper/scripts && mkdir -p out
curl -sL -o out/wikitext-2-raw-v1.zip \
    https://huggingface.co/datasets/ggml-org/ci/resolve/main/wikitext-2-raw-v1.zip
unzip -qo out/wikitext-2-raw-v1.zip -d out
python3 make_prompts.py          # out/prompt_{1k,4k,8k,16k,32k}.txt from prompt_1439.txt
./run_all.sh                     # ~2 h: headline (5 runs), context (3 runs), ablation (3), profiles,
    PPL, MoE
./run_correctness.sh             # ~15 min: reference logits at 8 (prompt, step) points, compiles
    logits_dump
./run_eval.sh                    # ~40 min: downstream check
python3 analyze.py               # out/results.json and the tables
python3 figures.py               # ../figs/{context,ablation,breakdown}.pdf

```

The scripts anchor paths on their own location, stop on the first missing prerequisite or failed command, and write every raw log to `out/logs/`. In our runs three measurements hit disk-bound stalls after the page cache had been turned over by other model files (one `llama.cpp` 16k-context run at 1.6 s per token, one `DS4 Q8_0` prefill at 44 t/s, one `Q4_0` run at 0.5 t/s); they were rerun with a warm cache and the packaged logs hold the reruns. The GPU must be otherwise idle: `DS4` refuses to start while another `DS4` process holds `/tmp/ds4.lock`, and timing is not meaningful with a second model process on the GPU. The short prompt is

```

<|im_start|>user
Write a short paragraph about the ocean.<|im_end|>
<|im_start|>assistant
<think>

</think>

```

(20 tokens); the 1,439-token prompt is `paper/scripts/prompt_1439.txt`.

4. Switches and profilers

```

DS4_METAL_DECODE_STAGE_PROFILE=1 DS4_METAL_CB_TIMES=1 # per-stage GPU spans (stage_agg.py
    aggregates)
DS4_TOKEN_TIMING=1 # CLI: sample / emit / forward per token; per-forward ms
    when greedy
DS4_GLM_LOGIT_DUMP=f DS4_QWEN_LOGIT_DUMP_STEP=k # logits of step k (0 = prefill output)
DS4_QWEN_DISABLE_FUSED_SWIGLU=1 # fused gate/up -> two matvecs + SwiGLU
DS4_QWEN_DISABLE_MULTI_PROJ=1 # multi-weight proj -> one launch each
DS4_QWEN_DISABLE_FAST_MV=1 # vectorized single-weight mv -> ggml-style
DS4_METAL_Q4K_KSPLIT_NSQ=0 # K-split (FFN down) off
DS4_METAL_QWEN_FA_NWG=32 # fixed 32 split-KV workgroups
DS4_QWEN_DISABLE_FUSED_NORM=1 # residual+norm fusion off
DS4_METAL_Q4K_MULTI_NRO=2 # multi-weight kernel rows per simdgroup

```

The Metal System Trace captures of §4 were recorded and exported with:

```

xcrun xctrace record --template 'Metal System Trace' --time-limit 120s \
    --output ds4.trace --launch -- ./ds4 -m gguf/Qwen3.8-27B-UD-Q4_K_M.gguf -p "$PROMPT" -n 32 --temp
    0
xcrun xctrace export --input ds4.trace \
    --xpath '/trace-toc/run[@number="1"]/data/table[@schema="metal-gpu-intervals"]'
xcrun xctrace export --input ds4.trace \
    --xpath '/trace-toc/run[@number="1"]/data/table[@schema="metal-application-command-buffer-
    submissions"]'

```

GPU busy fraction and per-token GPU time come from the union of the GPU intervals in the decode window; CPU encode time per token from the submission table's encoder durations.